

MANUEL BUENROSTRO MACEDO

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EDUCATION

Embry-Riddle Aeronautical University — Daytona Beach, FL

Bachelor of Science in Aerospace Engineering, Rocket Propulsion

Expected Graduation: May 2028 | GPA: 3.0 / 4.0

Relevant Coursework: Engineering Fundamentals, Engineering Graphics (CATIA V5), MATLAB Programming, Calculus II & III, Physics, Chemistry and Lab

ENGINEERING EXPERIENCE

Engineering Physics Propulsion Lab (EPPL) — Embry-Riddle Aeronautical University

Undergraduate Research Assistant | 2024 – Present

- Collaborated in mechanical integration and iteration testing of lab hardware.
- Presented technical solutions to sponsors, visitors, and faculty at lab showcase events.
- Applied engineering design principles to create, revise, and improve mechanical parts in Autodesk Inventor.
- Gained firsthand experience in a professional research lab setting.

TECHNICAL PROJECTS

OpenArms – Open-source Robotic Arm — EPPL, ERAU | January 2025 - Present

- Team member in the design of an open-source robotic arm attachment for OpenMutt.
- Responsible for CAD modeling and developing modular end-effectors.

OpenMutt – Open-source Quadruped — ERAU | September 2024 – Present

- Designed and fabricated a pan-tilt sensor mount to extend quadruped functionality using 3D printing and Autodesk Inventor.
- Designed and fabricated a quick disconnect mounting system for interchangeable end-effectors extending quadruped functionality using 3D printing and Autodesk Inventor

VEX Robotics Competition – Team 7121E | March 2020 – May 2024

- Led 7-member team to 21 regional, 3 state, and 1 world championship appearances; won 7 design-based awards.
- Team leader, head designer; built award-winning robots with pneumatics and CAD; maintained detailed engineering documentation.

TECHNICAL & PROFESSIONAL SKILLS

Software & Tools: Autodesk Inventor, CATIA V5, MATLAB (basic), Python (beginner), 3D printing, laser cutting, CNC (beginner), Windows Ecosystem, Visual Studio Code (Beginner), Prusa Slicer, Bambu Studio, Cura.

Soft Skills:

- Leadership, teamwork, problem-solving, public speaking, technical presentations, technical report writing
- Experience with shop tools, and safety procedures.
- Languages: Fluent in English and Spanish